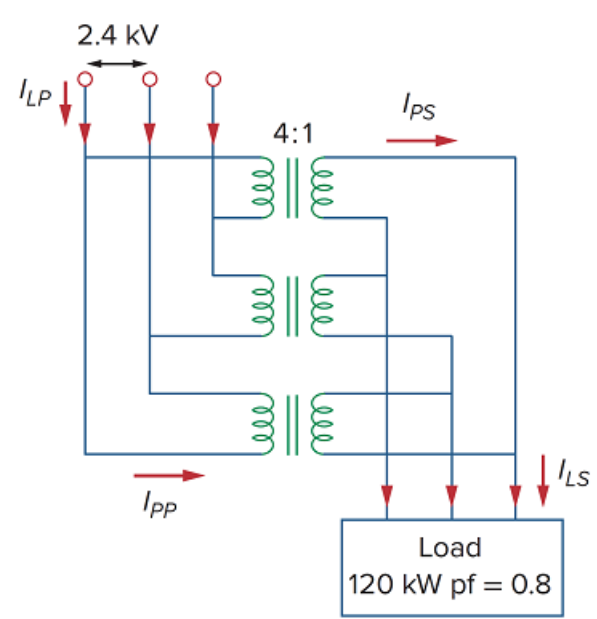


Problem

Consider the three-phase transformer shown in Fig. 13.136. The primary is fed by a three-phase source with line voltage of 2.4 kV rms, while the secondary supplies a three-phase 120-kW balanced load at pf of 0.8. Determine:

- (a) the type of transformer connections,
- (b) the values of I_{LS} and I_{PS} ,
- (c) the values of I_{LP} and I_{PP} ,
- (d) the kVA rating of each phase of the transformer.

Figure 13.136



Step-by-step solution

Step 1 of 6

(A) This is ? - ? connection.

Step 2 of 6

(B) The easy – way is to consider just one phase.

$$1 : n = a : 1$$

Or,

$$n = \frac{1}{4}$$

$$n = \frac{V_2}{V_1} \text{ which leads to}$$

$$V_2 = nV_1 = 0.25(2400) = 600$$

i.e.,

$$V_L P = 2400 V$$

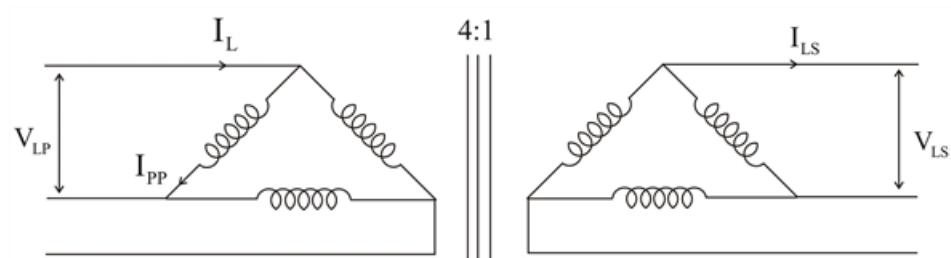
And,

$$V_{LS} = 60 V$$

$$S = \frac{P}{\cos \theta} = \frac{120}{0.8} KVA = 150 KVA$$

$$P_L = \frac{P}{3} = \frac{120}{3} = 40 KW .$$

Step 3 of 6



Step 4 of 6

But,

$$P_{LS} = V_{PS} I_{PS}$$

For the D – load,

$$I_L = \sqrt{3} I_p \text{ And } V_L = V_p$$

Hence,

$$I_{PS} = \frac{40000}{600} = 66.67 A$$

$$I_{LS} = \sqrt{3} I_{PS} = \sqrt{3} \times 66.67 = 115.48 A .$$

Step 5 of 6

(C) Similarly for the primary side,

$$P_{pp} = V_{pp}$$

Or,

$$I_{pp} = \frac{40000}{2400} = 16.667 \text{ A}$$

And,

$$I_{LP} = \sqrt{3} LP = 28.87 \text{ A} .$$

Step 6 of 6

(D) Since S = 150 K V A

Therefore,

$$S_p = \frac{S}{3} = 50 \text{ KVA} .$$